Enterprise Infrastructure Planning

NovaTech Startup Solutions

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For this project, I planned the IT setup of NovaTech, a fictional software company with 20 employees. The company needs computers, shared storage, internet access and a secure office network. The plan explains the equipment and software choices, how the devices connect, and how IT staff would maintain them. It is a design to review before buying or installing equipment.

Part 1 Company Profile

Company name: NovaTech Startup Solutions

Nature of business: A fictional software development startup building web and mobile applications for small businesses.

Vision: To become a trusted software solutions provider that helps small businesses grow through technology.

Mission: To deliver reliable, secure, and affordable software while developing employee skills and protecting customer information.

Office location: 3rd Floor, Meridian Business Center, Brgy. Real, Calamba City, Laguna (fictional).

Office arrangement: One office floor with departmental work areas and a locked equipment area. The CEO and department heads are included in the 20 employees, not additional positions.

Organizational structure

CEO / General Manager → IT Manager, HR Manager, Finance Manager and Sales Manager → staff within each department. Management roles are included in the headcounts below.

Department	Employees	Workstations
Information Technology	5	5 desktops
Human Resources	4	4 desktops
Finance	5	5 desktops
Sales	6	6 laptops
Total	20	14 desktops and 6 laptops

Enterprise Hardware Inventory

Each employee gets one workstation. Shared equipment serves the whole office. The asset IDs below group the planned purchases; each unit should receive its own asset record and serial number when delivered.

Asset ID	Hardware	Qty	Department	Purpose and planned specification
HW-001	Desktop computer	5	IT	8-core class CPU, 32 GB RAM, 1 TB SSD; development, testing and administration.
HW-002	Desktop computer	4	HR	6-core class CPU, 16 GB RAM, 512 GB SSD; employee records and office work.
HW-003	Desktop computer	5	Finance	6-core class CPU, 16 GB RAM, 512 GB SSD; accounting and financial reports.
HW-004	Laptop	6	Sales	16 GB RAM, 512 GB SSD, Wi-Fi and full-disk encryption; client visits and mobile work.
HW-005	Tower server	1	Shared / IT	8 cores, 32 GB ECC RAM, mirrored boot SSDs and mirrored data disks; files, Git and internal apps.
HW-006	VPN-capable router	1	Shared	WAN routing and provider handoff; avoid duplicate NAT with the firewall.
HW-007	Managed 24-port switch	2	Shared	Gigabit VLAN-capable core SW1 and access SW2; SW1 supplies PoE for two APs.
HW-008	Multifunction printer	2	Shared	One for IT/HR and one for Finance/Sales; network print and scan.
HW-009	UPS	3	Shared	Server/NAS, network edge/core, and access switch/AP path; size from measured load.
HW-010	Wireless access point	2	Shared	Business dual-band Wi-Fi, multiple SSIDs and VLAN tagging; coverage verified on site.
HW-011	NAS storage	1	Shared / IT	Two-bay mirrored storage, at least 4 TB usable initially; restricted local backups.
HW-012	External backup drive	2	IT	Encrypted 4 TB drives rotated weekly; one disconnected and stored off-site.
HW-013	Monitor	16	IT / HR / Finance	14 primary monitors plus two extra displays for IT development and administration.

I chose desktops for IT, HR and Finance because their work is mainly desk-based. Sales gets laptops for mobility. The 16 monitors cover the 14 desktops and two extra IT displays. Three UPS units protect the server, storage and network equipment. Items repeated in the network inventory refer to the same purchases. UPS ratings and backup storage must be checked against power use and data growth.

Enterprise Software Inventory

The software below supports office work, development and IT support. Use supported releases and record their exact builds during installation. Windows Pro licenses are assigned to devices, while Microsoft 365 subscriptions are assigned to users.

Software	Version	License	Purpose
Windows 11 Pro	Supported release	20 OEM/device licenses	Business desktop OS, disk encryption and workstation administration.
Ubuntu Server	24.04 LTS	Open source	Server OS for Samba file shares, Git access, SSH and internal applications. [1,2]
Microsoft Office	Microsoft 365 Apps	20 Business Premium user subscriptions	Word, Excel, PowerPoint and Outlook; business email and collaboration. [3]
VS Code	Approved stable build	Microsoft distribution license; no purchase fee	Code editing, debugging and development extensions for IT.
Git	Approved stable build	Open source (GPLv2)	Version history and collaboration for application code and configuration.
GitHub Desktop	Approved stable build	Open source (MIT)	Graphical Git workflow for IT; private repository access follows company policy.
VirtualBox	Supported 7.x base package	Open-source base package	Isolated development/test VMs on IT PCs. Optional Extension Pack excluded. [4]
Google Chrome	Approved stable build	Proprietary; no purchase fee	Browser-based business services and web application compatibility tests.
Microsoft Defender	Updated signatures / agent	Built-in AV; business management via Premium	Endpoint antivirus with centralized visibility using Defender for Business and Intune. [3]
AnyDesk	Supported business build	Paid commercial license	Authorized remote support by IT; no personal-use license for company support. [5]
7-Zip	Approved stable build	Open source; component licenses apply	Archive and extract files for routine document and backup transfers.

License planning: allocate one Windows Pro license per workstation and one Microsoft 365 subscription per employee. Scope AnyDesk licensing to the IT support team under the selected commercial plan. Review VirtualBox Extension Pack terms separately if its features are later required. Maintain a license register with purchase, assignment and renewal records. [3-5]

Enterprise Network Inventory

Item	Specification	Qty	Purpose
ISP modem / ONT	Provider-managed fiber handoff	1	Business fiber termination; bridge/routed handoff coordinated with ISP.
Router	Gigabit WAN/LAN and VPN capability	1	Provider-facing routing; firewall handles LAN policy, NAT and remote VPN.
Firewall	NGFW with VLAN interfaces, VPN and logging	1	Default-deny inter-VLAN rules, outbound control and inspected remote access.
Managed switch	24-port Gigabit, 802.1Q; PoE on SW1	2	Core SW1 and access SW2; VLAN trunks and wired endpoint connections.
Wireless AP	Dual-band, VLAN-aware business AP	2	Sales and guest coverage; PoE-powered from SW1.
Patch panel	24-port Cat6	1	Terminates 20 planned horizontal office cable runs; four spare positions.
Cat6 cable / patch lead	Correctly rated runs and patch leads	70	20 fixed runs, 20 cabinet leads, 18 endpoint leads, 6 device links and 6 spares.
RJ45 connector	Cat6-compatible plug; match cable type	100	Spare stock for patch-lead fabrication and repairs, not mandatory use on every run.

VLAN and addressing plan

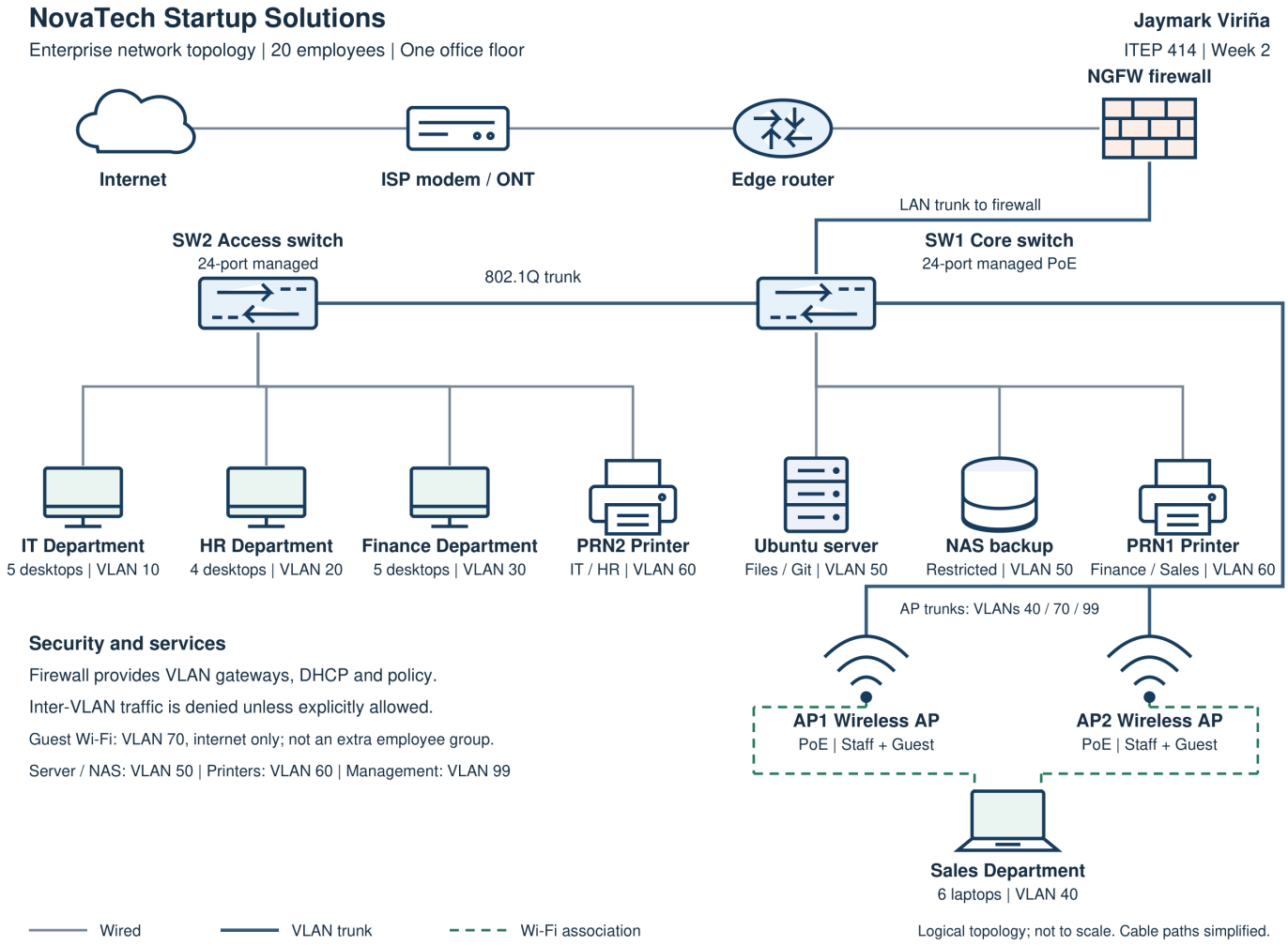
The firewall is the gateway at .1 in each subnet and serves departmental DHCP. Use .100-.199 for user leases; reserve infrastructure addresses. VLAN 99 manages devices and is not a normal user network.

VLAN	Group	Subnet	Intended access / addressing
10	IT	192.168.10.0/24	5 PCs; privileged access only from approved admin devices.
20	HR	192.168.20.0/24	4 PCs; access only to approved HR shares and services.
30	Finance	192.168.30.0/24	5 PCs; restricted accounting and finance shares.
40	Sales	192.168.40.0/24	6 laptops on staff Wi-Fi; approved files and printers.
50	Servers / backup	192.168.50.0/24	Server .10 and NAS .20; backup administration restricted.
60	Printers	192.168.60.0/24	PRN1 .11 and PRN2 .12; print access without broad LAN reach.
70	Guest Wi-Fi	192.168.70.0/24	Internet only with client isolation; blocked from private subnets.
99	Management	192.168.99.0/24	SW1 .11, SW2 .12, AP1 .21, AP2 .22; IT admin access only.

Access policy: permit approved departmental shares and printer services; deny direct department-to-department access. Only approved IT admin devices may manage infrastructure. Guest traffic can reach the internet only. SW1-SW2 and firewall-SW1 are tagged trunks; AP uplinks carry VLANs 40, 70 and 99. Restrict the NAS to backup/admin flows. Terminate fixed Cat6 runs on jacks and the patch panel; use connector types appropriate to the cable.

Enterprise Network Diagram

The diagram shows how the four departments reach the internet and shared services. Its two switches, two printers and two APs match the inventory. The server, NAS, printers and APs each have a switch connection. Guest Wi-Fi is separate from the employee network.



Connection and capacity notes

SW1 connects to the firewall, SW2, server, NAS, PRN1 and two APs: 7 used ports. SW2 connects 14 desktops, PRN2 and its uplink: 16 used ports. Sales laptops use Wi-Fi rather than adding six required wired endpoints. The two APs provide coverage for the same six laptops; their connections show possible association rather than simultaneous use of both APs.

The edge router handles provider routing and the firewall handles LAN NAT, security and remote VPN. Coordinate a routed handoff or bridge/pass-through arrangement with the ISP to avoid an accidental double-NAT design. Physical cable routes, wireless placement and RF capacity require an office survey before installation.

Editable source: diagrams/NovaTech_Network_Diagram.drawio. Image and print copies are provided as PNG and PDF. The patch panel is passive infrastructure and is omitted from this logical topology. [13]

System Administration Roles

Helpdesk Technician

Employees usually contact the helpdesk first when something stops working. At NovaTech, this could be a login problem, a printer error or a laptop that cannot connect to Wi-Fi. The helpdesk checks the problem and passes it on with useful details if another administrator is needed. [6]

Responsibilities: Log support requests and decide which need attention first. Set up employee devices, install approved software and check common account, printer and connection problems. Verify the employee's identity before a password reset. Record the checks already done before escalating a ticket.

Skills: Listening, clear explanations, Windows troubleshooting, basic hardware and networking knowledge, safe remote support and careful ticket notes.

Common tools: A ticketing system such as Jira Service Management, licensed AnyDesk, Windows Event Viewer, Intune, ping and ipconfig.

Certifications: CompTIA A+ is one option for building support skills. The Google IT Support Certificate is another starting point for troubleshooting, operating systems and customer support. [6]

Network Administrator

The Network Administrator looks after the connections between employees, shared devices and the internet. Cisco describes the role as installing, maintaining, monitoring and troubleshooting networks. At NovaTech, the switches, firewall and Wi-Fi would be part of this work. [7]

Responsibilities: Set up the router, firewall, switches and APs. Maintain VLANs, DHCP settings, DNS forwarding and access rules. Check link failures and Wi-Fi problems, monitor the network and keep backup copies of device configurations.

Skills: TCP/IP, subnetting, VLANs, routing, firewall rules, Wi-Fi planning and step-by-step troubleshooting.

Common tools: Device web interfaces and command lines, Wireshark for packet checks, an SNMP monitoring tool and Cisco Packet Tracer for practice and design exercises.

Certifications: Cisco CCNA covers network fundamentals, IP connectivity, security and automation. CompTIA Network+ is another foundation to consider. [8]

Linux and Cloud Administration

Linux System Administrator

The Linux System Administrator would maintain NovaTech's Ubuntu server, including shared files, user access and internal services. Ubuntu documentation and the RHCSA skill areas cover the user, storage, service and security tasks needed here. [1,9]

Responsibilities: Create users and groups, set file permissions and maintain Samba and Git access. Apply updates, check logs and storage space, schedule backups and test restores. Keep recovery instructions that another IT staff member can follow.

Skills: Linux commands, Bash scripting, file permissions, SSH, service management, storage and backup troubleshooting.

Common tools: OpenSSH, Bash, systemctl, journalctl, apt, rsync and systemd timers. Backup credentials should be kept separate from ordinary employee accounts.

Certifications: Red Hat Certified System Administrator (RHCSA) tests practical Linux administration. Its exam uses Red Hat Enterprise Linux, so Ubuntu-specific commands and tools still need separate practice. [9]

Cloud Administrator

The Cloud Administrator handles the company's hosted accounts and services. For this plan, that starts with Microsoft 365 and could later include cloud backup or hosting. Microsoft's Azure Administrator role also covers compute, networks, storage, identity and governance. [10]

Responsibilities: Manage cloud accounts and access, require MFA and review service health. Check backup storage, subscriptions and spending. Work with the Linux and network administrators when adding hosted services or moving an existing service to the cloud.

Skills: Cloud service models, account permissions, virtual networks, backup retention, cost monitoring and incident coordination.

Common tools: Microsoft 365 admin center and Entra admin center for the planned subscriptions. Azure portal, Azure CLI and budget alerts would be used if Azure resources are added.

Certifications: Microsoft Certified: Azure Administrator Associate is relevant to Azure infrastructure. Microsoft 365 administration would still require training in its own products and tools. [10]

How the four professionals work together

For example, an employee may report that a shared folder will not open. The Helpdesk Technician first checks the account and connection, then records the results in the ticket. The Network Administrator checks the network path or VLAN rule, while the Linux System Administrator checks the share and file permissions. The Cloud Administrator joins in if cloud identity or a hosted service is involved. They keep their notes in the same ticket and confirm with the employee that access works again. With only five IT employees at NovaTech, one person may handle more than one of these roles; the responsibilities still need to be clear.

Infrastructure Recommendations

Internet provider

I would ask PLDT Enterprise and Converge Business for business-fiber proposals for the office location. A 300 Mbps connection is a starting point for the plan, but the choice should depend on actual coverage, upload speed, repair response, uptime commitments and cost. A static IP is worth checking if a service needs one. Large backups can run after office hours. A separate mobile connection can provide failover if the main line goes down. Coverage, prices and service terms still need confirmation.

Server specifications

An 8-core CPU and 32 GB ECC RAM should be a reasonable starting point for shared files, Git and small internal applications. I would use two 480 GB SSDs in a mirror for the OS and two 4 TB disks in a mirror for data, with room to upgrade memory. Ubuntu Server 24.04 LTS offers a stable support lifecycle. Storage and workload measurements should guide later upgrades. Mirrored disks help when one drive fails, but they do not replace a backup. [1,2]

Backup strategy

Keep the working files on the server and make a versioned backup to the NAS each night. Copy them weekly to encrypted rotating USB drives, keeping one disconnected and off-site. An encrypted cloud copy of critical files would reduce the gap between off-site backups. The initial targets are to lose no more than 24 hours of data and restore service within 8 hours; a restore trial must show whether these targets are realistic. Without the cloud copy, an off-site recovery could lose up to one week. Keep 30 daily and 12 monthly versions where storage allows. Check failures daily, restore sample files monthly and test full recovery quarterly. [11]

Security and Growth

Security recommendations

Use the VLAN rules in Part 4 so employees can reach the files and printers they need without gaining access to every department. Allow SSH administration only from approved IT devices, use keys and disable direct root login. Remote administrators should connect through a VPN with MFA. Keep SSH and file shares off the public internet. Encrypt employee devices, apply updates, limit administrator rights, lock the equipment area and back up device configurations. Any inspection of encrypted traffic needs an approved company policy.

Antivirus

Keep Microsoft Defender Antivirus active on all Windows workstations. This plan includes Microsoft 365 Business Premium for 20 users, covering Office apps, Intune management and Defender for Business. IT still needs to enroll the devices and test whether alerts reach the administrator. When a threat is detected, isolate the device, investigate the cause and record the recovery steps. [3]

Password policy

Require unique passphrases of at least 15 characters and MFA for company email, cloud administration and VPN access. A company password manager can help employees avoid reusing passwords. Block known compromised passwords and change credentials when compromise is suspected. Avoid routine forced changes and arbitrary character rules. Keep administrator accounts separate from everyday accounts and protect recovery methods. These recommendations follow NIST guidance and need to be configured within each product's available controls. [12]

Expansion plan

SW1 uses 7 ports and SW2 uses 16, leaving 25 of the 48 ports free. The 14 desktops use cable connections and the six Sales laptops use Wi-Fi. The 20 fixed cable runs cover the desktops, two printers, two APs and two spare outlets. Before adding staff, check the PoE budget, Wi-Fi load, patch-panel capacity and server use. Add another panel, switch or AP when those checks show it is needed. As the company takes on more work, review whether a second connection, more resilient equipment or cloud hosting is worth the cost.

Personal Reflection

I used to think that a System Administrator mainly fixes computers and installs software. Working on this plan for NovaTech showed me how much needs to be decided before any equipment is bought. The company has 20 employees, but their needs are different. The plan has to account for the work they do, the information they use, and what happens if a device or service fails.

The hardware inventory made those differences easier to understand. IT, HR, and Finance can use desktops because they mainly work at their desks. Sales needs laptops for work that may involve visiting clients. Shared equipment also needs a reason for being included. A NAS, backup drives, and UPS units may be less noticeable than computers, but they help protect the company's files and keep important services available.

The network diagram was the hardest part for me. Listing devices was straightforward; showing how they connect took more thought. The drawing also had to agree with the inventory. For example, the 14 desktops use wired connections, while the six Sales laptops use Wi-Fi. I needed to understand that connecting devices to a switch does not automatically mean every department should have access to the others. That is where the VLANs and firewall rules come in.

This activity also changed how I look at backups. Keeping another copy of a file is only part of the job. Someone still has to check whether the backup ran and whether the files can actually be restored. Software licenses and spare network ports also need attention early, since overlooking them can lead to extra costs after installation.

Planning gives the company something to check before spending money. It also gives the people setting up the network a clear starting point. For me, this project is practice in explaining my choices instead of just naming equipment. Learning about the helpdesk, network, Linux, and cloud roles showed me how their work connects during a problem. I still need hands-on experience configuring these systems, but I now have a better idea of what to check and document before deployment.

Reflection length: approximately 346 words.

References

I used these sources for the role descriptions, software licensing and security recommendations. Equipment quantities and access rules are my design choices for the fictional company. Sources accessed: 25 October 2026.

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